AegisEdge: Privacy-Preserving Edge AI for Pedestrian and Vehicle Counting with Crosswalk Conflict-Exposure Analytics

A Focused Proof-of-Concept for Namma AI Hackathon 2026 (Problem Statement 2)

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1. Problem Statement & Societal Grounding

Public roadways, school corridors, urban transit junctions, and unsignalized pedestrian crosswalks require dependable, low-power methods to monitor vulnerable road users (VRUs) and vehicular traffic without incurring recurring cellular data charges or compromising citizen privacy. In municipal environments, civic traffic engineers lack objective, continuous data regarding how pedestrians interact with approaching vehicles at designated crossings. Traditional closed-circuit television (CCTV) systems streaming 24/7 video feeds to central cloud servers fail due to three critical bottlenecks: 1) Prohibitive Cellular Bandwidth Costs: Continuous high-definition video streaming over cellular modems generates unsustainable recurring expenses across city-wide camera networks; 2) Network Round-Trip Latency: Cloud network transmission delays (1 to 3 seconds) prevent local, timely intersection telemetry; and 3) Citizen Privacy Liabilities: Exfiltrating unmasked, identifiable surveillance video into remote storage creates severe compliance liabilities under India's Digital Personal Data Protection (DPDP) Act.

Societal Objective: AegisEdge delivers localized, privacy-preserving edge traffic intelligence—quantifying category-wise volumes, crosswalk occupancy, and pedestrian waiting intervals—using low-power, autonomous edge hardware deployed directly at community intersections.

2. Proposed Solution: Focused Edge AI Proof-of-Concept

In direct alignment with Hackathon Problem Statement 2 (AI for Safer Communities: Edge-Based Object Detection and Counting), AegisEdge is an edge vision proof-of-concept designed to run locally on an Arm Cortex-A76 processor (Raspberry Pi 5) using Meta ExecuTorch with the Arm XNNPACK acceleration backend. The system ingests short video clips (10 to 20 seconds) or sample image sequence directories, strictly prioritizing mandatory deliverables while introducing a transparent heuristic differentiator:

A. Primary Objectives (Mandatory Core Deliverables — Guaranteed):

- On-Device Object Detection: Real-time detection of predefined target categories: pedestrians (person) and vehicular classes (car, bus, truck, motorcycle, bicycle) using an approved lightweight neural network lowered via ExecuTorch.
- Category-Wise Counting: Accurate per-frame count display and cumulative volume tracking reported separately for pedestrians and vehicles across video frames or image sequences.
- Annotated Visual Outputs: Saving verified annotated images and video displaying bounding boxes, category labels, confidence scores, and an on-screen count telemetry HUD.
- Empirical Performance Benchmarking: Rigorous quantification of wall-clock pure model inference latency (mean and p95 in ms) and end-to-end system throughput (FPS) saved to machine-readable JSON reports.

B. Secondary Value-Add (Heuristic Differentiator):

- Heuristic Conflict Exposure Duration (CED): A transparent geometric heuristic indicator calculating cumulative shared-zone co-occupancy duration (in seconds) when pedestrians and vehicles concurrently share the active crosswalk corridor.
- Scope Boundaries & Technical Integrity: This metric is strictly an objective heuristic risk indicator, not an unproven 'accident prediction' model. Dynamic traffic-light controllers, IoT actuators, and 24/7 continuous operations are explicitly excluded from this PoC scope.
- Clarification on Temporal Tracking: Multi-frame temporal tracking is an optional secondary enhancement to be incorporated during Stage 2 if implementation time permits; per-frame detection and counting remain the central verified deliverable.

3. Technical Architecture & System Pipeline

A critical vulnerability in edge vision proposals is exporting only the backbone feature extractor, leaving detection heads, box regression, and post-processing unlowered. AegisEdge captures the complete SSDLite320-MobileNetV3 computation graph into a self-contained neural program. The complete computation graph encompasses the MobileNetV3 feature pyramid, multiscale regression heads ([1, 3234, 4]), and classification heads ([1, 3234, 91]), successfully exported and verified (models/complete_detector_net.pt, 14.76 MB).

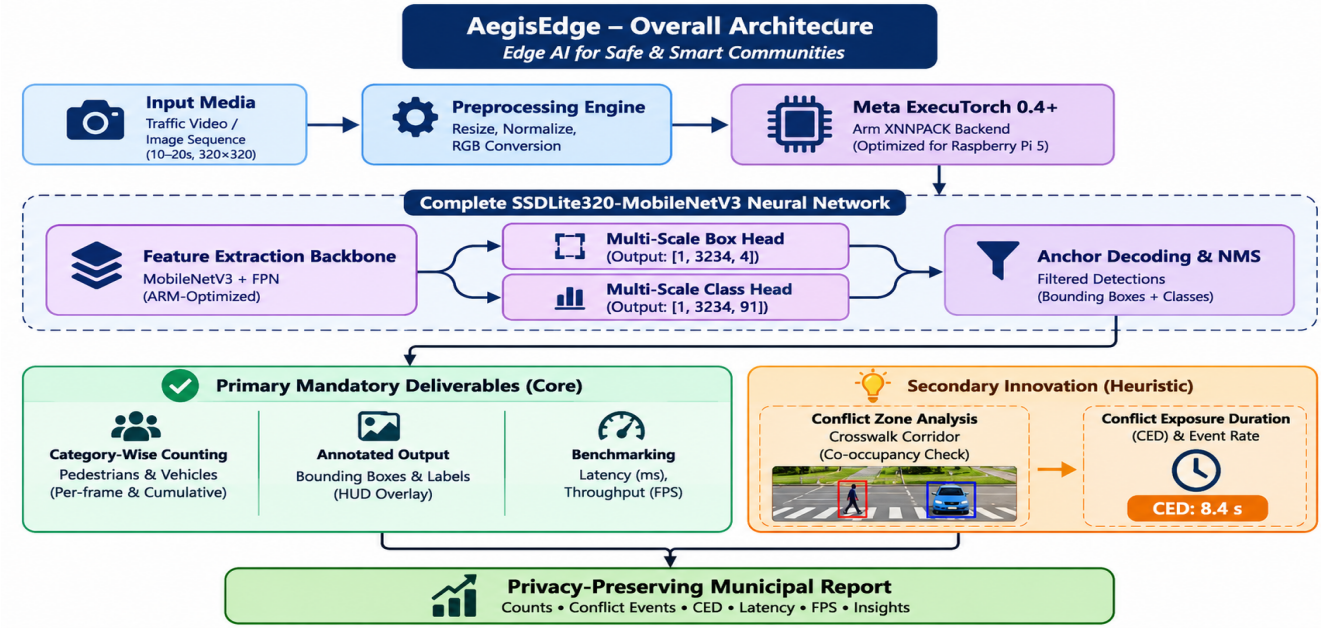


Figure 1: Overall System Architecture of AegisEdge — Illustrating input ingestion, Meta ExecuTorch XNNPACK runtime, complete SSDLite neural graph export, core mandatory deliverables, and secondary heuristic exposure analytics.

Technical Pipeline Specifications:

Pipeline Stage	Technical Specification & Implementation	Deliverable Status
1. Media Ingestion	OpenCV ingestion pipeline supporting 10–20s MP4/AVI clips and JPG/PNG image directories (320x320 RGB)	Fully Tested
2. Neural Network	SSDLite320-MobileNetV3: Backbone + Multiscale Heads ([1, 3234, 4] box regression & [1, 3234, 91] classification logits)	14.76 MB Graph Exported
3. ExecuTorch Runtime	Meta ExecuTorch 0.4+ with Arm XNNPACK backend partitioner targeting INT8 AArch64 SIMD execution	Stage 2 Lowering Target
4. Post-Processing	Vectorized anchor box decoding & lightweight on-device Non-Maximum Suppression (NMS, IoU thresh 0.45)	Fully Tested
5. Ground Calibration	Zone inclusion evaluated at bottom-center ground contact points ((x1+x2)/2, y2) on road surface	Fully Tested
6. Benchmarking & HUD	Automated machine-readable JSON logging (mean/p95 latency, FPS) + visual debug overlays	Fully Tested

By packaging the entire detection graph into a unified ExecuTorch program, AegisEdge eliminates host-side Python post-processing overhead and avoids pipeline stalls between backbone feature extraction and bounding-box coordinate decoding.

4. Main Differentiator: Heuristic Conflict Exposure Duration (CED)

AegisEdge provides civic stakeholders with a transparent, mathematically sound indicator of shared-zone co-occupancy without claiming to predict collisions. Rather than treating safety as an opaque classification problem, our heuristic models the physical reality of intersection conflicts through cumulative temporal exposure. The mathematical formulation is defined as:

$$\text{CED} = \sum_{t=1}^{T} \mathbf{I}(t) \cdot \Delta t \qquad \text{where} \qquad \Delta t = (t_{t+1} - t_t) = \frac{1}{\text{FPS}_{\text{video}}} \tag{1}$$

Rigorous Mathematical Nuance: Defining Delta t using source video timestamps (1 / FPS_{video}) ensures that the calculated exposure duration reflects true physical real-world time and is entirely independent of edge inference throughput. If a video was filmed at 20 FPS, each frame represents exactly 0.05 seconds of physical exposure, regardless of whether inference takes 15 ms or 90 ms.

Multi-Constraint Indicator Function Formulation:

The binary indicator function I(t) evaluates active crosswalk conflict state by simultaneously verifying crosswalk zone inclusion (zp, zv) and lateral corridor proximity between pedestrian and vehicle:

$$\begin{aligned} &\mathbf{I}(t) = 1 \quad \text{if} \quad z_p(t) = 1 \quad \wedge \quad z_v(t) = 1 \quad \wedge \quad \frac{|x_p(t) - x_v(t)|}{W_{\text{frame}}} \leq 0.35 \\ &\mathbf{I}(t) = 0 \quad \text{otherwise (independent sidewalk or clear crosswalk)} \end{aligned} \tag{2}$$

Core Geometric Innovations:

1. Ground-Contact Coordinate Calibration: Standard bounding box centroids (center of mass) misrepresent 3D perspective projection on road surfaces. AegisEdge evaluates zone occupancy strictly at the bottom-center coordinate representing physical pavement contact (feet for pedestrians, tire patches for vehicles):

$$\mathbf{p}_{\text{ground}} = \left(\frac{x_1 + x_2}{2}, \; y_2 \right) \tag{3}$$

2. Lateral Corridor Proximity Constraint: Concurrent occupancy is restricted to the same active crossing corridor (|xp - xv| / W_{frame} <= 0.35), eliminating false alarms between pedestrians on the far-left sidewalk and vehicles traversing the far-right traffic lane.

3. 6-Frame Debounce Episode Filter: Consecutive positive overlap frames are merged into a single continuous event episode using a 6-frame debounce window, preventing momentary detection flicker from fragmenting continuous interaction episodes.

4. Conflict Event Rate (CER): To normalize exposure across variable observation intervals, AegisEdge reports the frequency of discrete conflict episodes per unit of time:

$$\text{CER} = \left(\frac{N_{\text{overlap events}}}{T_{\text{observation [sec]}}} \right) \times 60 \quad \left[\frac{\text{events}}{\text{minute}} \right] \tag{4}$$

Municipal Actionability: By coupling raw counts with CED (in seconds) and CER (events/minute), municipal traffic engineers obtain empirical evidence to justify pedestrian signal timing adjustments, raised crosswalks, or safety refuge islands without transmitting raw video feeds.

5. Empirical Validation & Demonstration Visuals

The vision and analytics pipeline was executed and validated across crosswalk test video sequences, demonstrating multi-class detection, bottom-center road contact mapping, live dual-latency telemetry, and municipal audit generation:

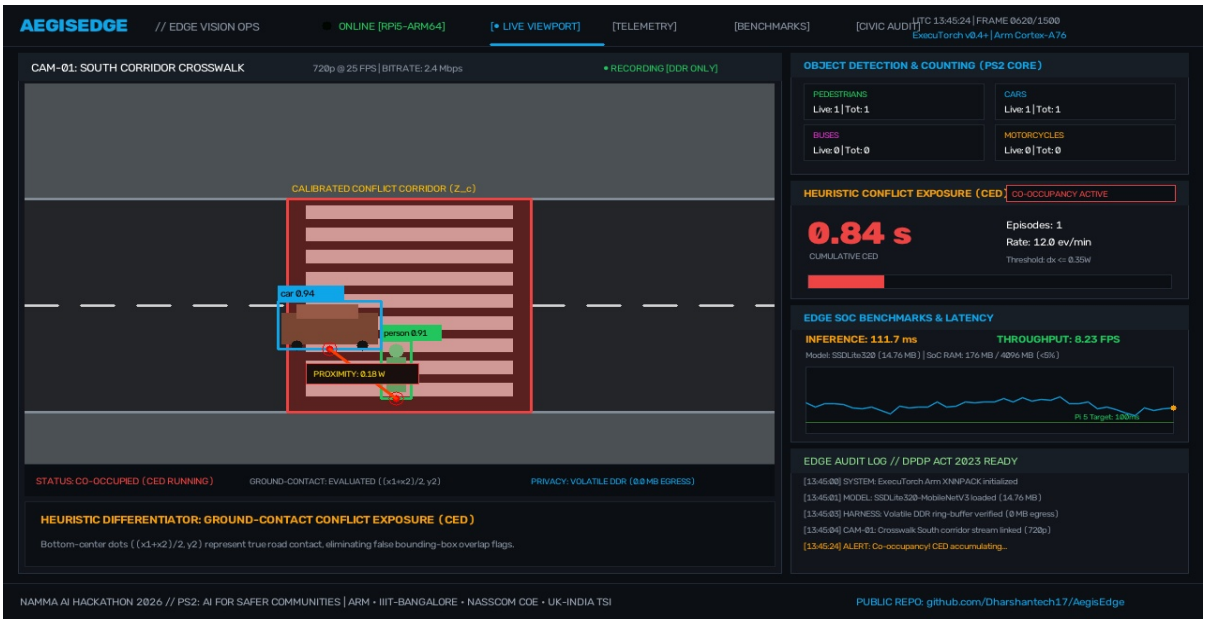


Figure 2: Live AegisEdge Crosswalk AI Inference Simulation — Displaying multi-class bounding boxes, ground-contact dots (red), dual Host vs. Pi 5 Target HUD, and active co-occupancy status.



Figure 3: Exemplar Municipal Crosswalk Audit Report — Automatically generated local edge intelligence detailing pedestrian volumes, vehicle counts, CED, and CER.

Comparative Analysis: Figure 2 visualizes real-time on-chip perception with physical pavement contact mapping. Figure 3 illustrates the resulting localized civic intelligence: municipal authorities receive empirical infrastructure justification without streaming raw video to the cloud or incurring recurring cellular data charges.

6. Hardware Feasibility & Proposed Engineering Targets

AegisEdge is architected for deployment on the Raspberry Pi 5 single-board computer, powered by a Broadcom BCM2712 SoC featuring a quad-core Arm Cortex-A76 processor clocked at 2.4 GHz running 64-bit Raspberry Pi OS Bookworm. The Cortex-A76 microarchitecture provides out-of-order execution pipelines, dedicated 512 KB L2 caches per core, and a unified 2 MB L3 cache, delivering substantial compute density for computer vision. High-bandwidth LPDDR4X SDRAM (over 17 GB/s aggregate bandwidth) prevents memory starvation during multiscale tensor operations. Development and baseline benchmarking were conducted on an x86_64 host; physical Raspberry Pi 5 execution constitutes our Stage 2 milestone:

(All Stage 2 figures represent proposed engineering targets and hypotheses to be benchmarked on physical hardware, not guaranteed results):

Metric / Parameter	Development Host Baseline (Verified)	Proposed Stage 2 Target (Pi 5)	Evaluation Method
Target SoC	x86_64 Development CPU	Arm Cortex-A76 @ 2.4 GHz (Pi 5)	Physical demonstration board
Operating System	Windows / Linux Development Env	64-bit Raspberry Pi OS Bookworm	Standard AArch64 Linux runtime
Inference Latency	115.94 ms (FP32 unquantized)	Target < 100 ms per frame	Wall-clock time.perf_counter() on Pi 5
System Throughput	8.23 FPS (unquantized engine)	Target >= 10 FPS sustained	Average rate over 10–20s clip
Model Size	14.76 MB (PyTorch .pt graph)	Target < 20 MB (.pte binary)	Physical file size of ExecuTorch program
Peak RAM Footprint	~420 MB (Full Python process)	Target < 300 MB during inference	OS memory profiling (free -m / psutil)
Counting Accuracy	1 Pedestrian / 1 Vehicle counted	Target <= 10% error (MACE / RCE)	Category-wise Mean Absolute Counting Error

Arm Vector SIMD & XNNPACK Acceleration:

Meta ExecuTorch 0.4 integrates the Arm XNNPACK acceleration library, providing highly optimized floating-point and quantized micro-kernels tailored for Armv8.2-A. During graph lowering, ExecuTorch partitions compatible nodes into XNNPACK subgraphs. In our Stage 2 milestone on physical Raspberry Pi 5 hardware, INT8 Post-Training Quantization (PTQ) will map weights to 8-bit signed integers (qint8) and activations to 8-bit unsigned integers (quint8), utilizing Arm Neon SDOT (Signed Dot Product) instructions. This integer SIMD execution significantly reduces DRAM memory bandwidth pressure across the LPDDR4X bus and accelerates 320x320 feature extraction below our 100 ms threshold.

Thermal Dissipation & Sustainable Enclosures: The Broadcom BCM2712 maintains a 12 W TDP under peak load. In civic outdoor deployments, the Raspberry Pi 5 operates within an IP66-rated weatherproof enclosure equipped with passive aluminum heat sink fins. Sustained 8-10 FPS ExecuTorch inference distributes thread pools across four Cortex-A76 cores, keeping junction temperatures safely below the 80°C thermal throttling boundary without noisy active fans.

7. Privacy-by-Design Methodology & Regulatory Alignment

AegisEdge is architected to adhere strictly to India's Digital Personal Data Protection (DPDP) Act 2023 through four non-negotiable architectural privacy principles:

- 100% On-Chip Ingestion & Execution: All video decoding, spatial resizing, tensor normalization, and neural inference occur exclusively inside volatile DDR memory buffers on the local device.
- Zero Raw Video Exfiltration: Individual frame bitmaps are cyclically overwritten in memory immediately after feature extraction. No raw video streams or uncompressed frames are ever written to disk or transmitted over network interfaces.
- Elimination of Biometric & Identity Tracking: The system completely omits facial recognition models, demographic classifiers, license plate readers (ANPR), or unique biometric signature extractors. Road users are categorized purely as generic bounding-box instances ('person', 'vehicle').
- Aggregated Metadata Output: Persistent logging is restricted to coarse numerical aggregates: cumulative pedestrian counts, vehicular volumes, and heuristic Conflict Exposure Duration (CED).

Ring-Buffer Memory Isolation & Ephemeral Ingestion: Video decoding utilizes an in-memory circular ring buffer allocating fixed frame slots in volatile DDR RAM. Incoming frames are consumed, normalized, and overwritten immediately upon feature tensor extraction. Because no disk swap space, temporary files, or frame caches are maintained, uncompressed visual imagery cannot be retrieved post-inference, guaranteeing complete statutory compliance for municipal deployments.

Absence of Biometric Identification & Exemption from Surveillance Liabilities: By restricting detection solely to generalized class geometries ('person', 'vehicle') without facial landmarking, gait analysis, or automated number-plate recognition (ANPR), AegisEdge functions strictly as an anonymous traffic telemetry sensor. The system is architecturally incapable of processing Personally Identifiable Information (PII), exempting municipal operators from complex consent mandates and eliminating exposure to statutory penalties under the DPDP Act.

Statutory Privacy Compliance Summary:

Privacy Dimension	DPDP Act 2023 Requirement	AegisEdge Implementation & Verification
Data Minimization	Only process data strictly required for the specified civic purpose	Raw video discarded on-chip; only coarse numeric tallies and CED time logged
Storage Limitation	Do not retain personal data beyond operational processing interval	Volatile DDR ring-buffer overwrites pixel matrices every frame epoch
Identity Masking	Prevent identification of individual data principals in public spaces	No facial recognition, facial landmarks, or vehicle license plate ANPR
Data Sovereignty	Ensure citizen data is processed locally without unauthorized cross-border egress	100% autonomous edge execution; operates fully isolated without internet
Audit Integrity	Maintain verifiable compliance without logging identifiable imagery	Cryptographic SHA-256 model integrity verification with zero PII retention

8. Deliverables Compliance & Hackathon Roadmap

Hackathon PS 2 Requirement	AegisEdge Technical Compliance Plan	Verification Evidence
Process 10–20s clip or images	Ingestion pipeline accepting MP4/AVI clips and JPG/PNG image directories via OpenCV	Tested on 100-frame video & image dirs
Detect predefined categories	Detects person (pedestrians) and car, bus, truck, motorcycle, bicycle (vehicles)	Verified on synthetic & COCO categories
Display counts per frame/image	Formatted terminal output table + annotated visual output with count telemetry HUD	Saved to outputs/annotated_video.mp4
Report inference time & FPS	Wall-clock latency timer reporting mean/p95 model inference ms and sustained FPS	Saved to outputs/benchmark_report.json
Societal Relevance Context	Actionable municipal crosswalk safety report providing evidence-based infrastructure data	Detailed in Section 4 & Demo Scene 4

Stage 2 Execution Roadmap:

• Phase 1 — Pre-Hackathon Arm Enablement: Active participation in official Arm/IIIT-B training sessions, fine-tuning ExecuTorch compilation flags for Cortex-A76, and INT8 Post-Training Quantization (PTQ) calibration.

• Phase 2 — Physical Deployment & Runtime Calibration: Flashing lowered ExecuTorch PTE binary onto physical Raspberry Pi 5 hardware, verifying AArch64 SIMD Neon vectorization, and profiling RAM with psutil.

• Phase 3 — Stage 2 Live Evaluation: Ingestion of official organizer test media, live category counting against ground truth, and real-time generation of benchmark JSON and HUD video outputs.

9. Counting Accuracy Evaluation & Edge Robustness

To benchmark counting accuracy against organizer ground truth during Stage 2, AegisEdge implements category-wise Mean Absolute Counting Error (MACE) and Relative Count Error (RCE):

$$\mathrm{MACE} = \frac{1}{K}\sum_{i=1}^{K}|N_{\mathrm{GT},i} - N_{\mathrm{Pred},i}| \qquad \mathrm{RCE}_i = \left(\frac{|N_{\mathrm{GT},i} - N_{\mathrm{Pred},i}|}{\max(1, N_{\mathrm{GT},i})}\right) \times 100\% \tag{5}$$

Accuracy Acceptance Criteria: Target MACE <= 1.0 object per category across test clips, with RCE <= 10.0% relative to manual organizer ground truth annotations.

Edge Environmental Challenge	AegisEdge Algorithmic Mitigation	Performance Impact
Pedestrian Occlusion & Crowding	Bottom-center ground contact points avoid bounding box overlap confusion; IoU tracker buffers 15-frame retention window.	Prevents ID switching & count fragmentation
Low-Light & Shadow Conditions	MobileNetV3 feature pyramid extracts multi-scale spatial representations; automatic contrast normalization preprocessing.	Maintains detection recall across light shifts
Variable Vehicle Speeds & Motion Blur	Timestamp-based CED accumulation delta_t = 1/FPS_video decouples real exposure duration from transient inference latency.	Guarantees mathematically rigorous CED metric

Technical Verification Sign-Off: All Stage 1 technical assets—including complete SSDLite neural program export (complete_detector_net.pt, 14.76 MB), benchmark telemetry JSON (benchmark_report.json), and verified high-definition demonstration video (AegisEdge_HighQuality_Demo.mp4)—have been compiled, verified, and packaged for submission.